Vacation-rental marketplace — production architecture

Conceptual target for this assignment. Only the highlighted web-frontend slice exists in this repository; the rest is described in docs/ARCHITECTURE.md.

APPLICATION CLIENTS

Web application — IMPLEMENTED HERE
Next.js App Router · SSR/ISR listing pages
Static route "/" — this is the implemented part

Guest mobile apps
iOS and Android clients over HTTPS; native booking flows against the same public API

Host & partner tools
Listing management, calendars, pricing rules and payout dashboards for hosts

HTTPS

EDGE & CDN

Edge CDN & WAF
TLS termination, DDoS protection, static assets. HTML and listing JSON are cached at the edge keyed by listing, locale and currency: short TTL with stale-while-revalidate, immediate purge on content or price change. Only cache misses and API calls continue to the origin.

API calls

API GATEWAY

API gateway
Authentication and authorization (sessions, OAuth, JWT), rate limiting, request shaping and API versioning. Every domain service sits behind it — clients never address a service directly, so policies and quotas are enforced once, in one place.

routed per service

DOMAIN SERVICES

Listings & pricing
content, photos, rates, calendars

Search
query, ranking, facets, geo

Bookings & availability
holds, reservations, confirmations

Payments
charges, refunds, payout ledger

Auth & accounts
identities, sessions, OAuth

Reviews & content
ratings, questions, reports

Notifications
email, SMS, push fan-out

Media & imaging
upload, scan, image derivatives

ASYNC EVENTS

reads & writes

async publish

Event bus (queue) with dead-letter topic
booking.confirmed · listing.updated · payment.captured · search.reindex · notify.* — published by one service, consumed by any others.
Retries with backoff and a dead-letter topic, so slow consumers never block a user request.

media blobs

index updates

DATA & STORAGE

Primary database
PostgreSQL — bookings, inventory, users. Serializable transactions plus idempotency keys prevent double-booking.

Read replicas
Listing and reviews reads at scale; replicas absorb the read load in front of the primary.

Cache (Redis)
Hot listing JSON, sessions and availability, with explicit invalidation on every write.

Search index
Elasticsearch / OpenSearch: geo and availability filters, ranking, facets. Falls back to SQL when the index is stale.

Object storage + CDN
S3-compatible blobs and AVIF/WebP derivatives with immutable cache headers; the database stores only refs.

Read path: gateway → cache → replicas → primary. Write path: primary → invalidation → cache; domain events → queue → search index and notifications.

PLATFORM & OPS

CI/CD
lint → typecheck → tests → build → visual regression → staged rollout with a canary

Observability
Metrics, structured logs, distributed traces; SLOs on p95 latency and error rate

Security & config
Secrets, least-privilege service accounts, audit logs, feature flags

synchronous request / response · asynchronous event · implemented in this repository (web frontend only) · conceptual target (not implemented)